Vojtěch Sýkora

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Experience

Miton, AI Engineer & Product Owner — 10/2025 – 04/2025, Prague, Czechia, Hybrid

- Sole engineer and project lead on an AI-powered data intelligence platform - owned architecture, product decisions, and stakeholder communication
- Built embedding-based semantic search over 150K+ records (pgvector, HNSW, JSONB prefilters, quality reranking) and multi-source data ingestion, enrichment, and identity resolution pipelines
- Built the full backend (FastAPI, PostgreSQL, SQLAlchemy) with LLM-powered extraction, multi-environment deployment, auth, and admin tooling

Stealth Startup (AI & Robotics), Founder & CEO — 05/2025 – 01/2026, Munich, Germany, On-site

- Founded a robotics & AI startup building automated shelf-restocking for supermarkets - led investor pitching and secured LOIs with REWE Digital
- Trained object detection and instance segmentation models (YOLOv11) achieving 94.5% mAP50 on shelf product detection from a small initial dataset
- Designed a scalable dataset pipeline: DVC versioning, session-based metadata architecture, multi-format export (COCO/YOLO), built for international rollout

Synthavo, Agentic AI Engineer — 12/2024 – 07/2025, Stuttgart, Germany, Hybrid

- Built an agentic AI system for industrial spare part recognition - reduced identification from hours of manual lookup to seconds via conversational AI (LangChain, LangGraph, ChromaDB, RAG)
- Delivered a working prototype independently from concept to demo; company hired an external team to scale it into production

Education

University of Tuebingen, Master's in Machine Learning — 10/2023 – 09/2025, Tuebingen, Germany

- DAAD full scholarship (2 years). Thesis: Multimodal Deep Learning for Automated Schematic Analysis. Focus: deep learning, computer vision.

Czech Technical University in Prague, Bachelor's in Open Informatics (AI specialization) — 09/2020 – 06/2023, Prague, Czechia

- Data structures & algorithms, Python, C/C++, Java, SQL, GPU programming. prg.ai interdisciplinary AI minor. Thesis: Reinforcement Learning

Skills

Python, FastAPI, PostgreSQL, pgvector, LLMs, OpenAI, AI agents, RAG, LangChain, vector search, PyTorch, OpenCV, computer vision, Docker, VPS, Git, Linux

Claude Code (multi-agent orchestration, MCP servers), Codex, prompt engineering, stakeholder communication

English (Fluent), Czech (Native). German (Intermediate),